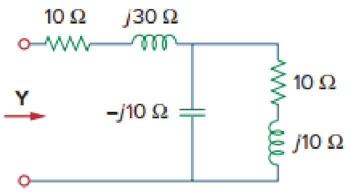
Chapter 9, Problem 37P

Problem

Determine the admittance **Y** for the circuit in Fig. 9.44.

Figure 9.44:



Step-by-step solution

Step 1 of 2

Refer to Figure 9.44 in the text book for the circuit.

Determine the impedance of the network.

$$\begin{aligned}\mathbf{Z} &= (10+j30)+(-j10)\,\|\,(10+j10)\\ &= (10+j30)+\frac{(-j10)\times(10+j10)}{(-j10)+(10+j10)}\\ &= 10+j30+10-j10\\ &= 20+j20\ \Omega\end{aligned}$$

Step 2 of 2

The admittance **Y** is defined as the reciprocal of the impedance.

Determine the admittance **Y.**

$$\begin{aligned}\mathbf{Y} &= \frac{1}{\mathbf{Z}}\\ &= \frac{1}{20+j20}\\ &= 0.025-j0.025\\ &= (25-j25)\times10^{-3}\ \mathrm{S}\end{aligned}$$

$$=(25-j25)\ \mathrm{mS}$$

Thus, the admittance $\mathbf{Y}$ of the circuit is $\boxed{(25-j25)\ \mathrm{mS}}$.